HANDWRITTEN NOTES

COMPLEX VARIABLES, PROBABILITY AND STATISTICS

Subject Code : 23A54403

UNIT - V

Estimation, Large Sample Tests and Confidence Intervals

Topics covered in this unit :

- Estimation: parameters, statistics, sampling distribution
- Point estimation
- Null and alternative hypothesis; critical and acceptance regions
- Level of significance, two types of errors, power of the test
- Large sample tests: single proportion, difference of proportions
- Large sample tests: single mean, difference of means
- Confidence intervals: one-sample and two-sample problems

(Key terms are written in red , headings in purple/green, formulas on yellow.)

1. Population, Sample, Parameter, Statistic

Term	Meaning
Population	Complete collection of items under study (size N)
Sample	Part of the population actually observed (size n)
Random sampling	Every unit has equal chance of selection (with / without replacement)
Parameter	Numerical characteristic of the population (mu, sigma, P) - constants
Statistic	Function of sample values (x-bar, s, p) - random variables
Large sample	n >= 30; small sample: n < 30

	Population (parameter)	Sample (statistic)
Mean	mu	x-bar
Standard deviation	sigma	s
Proportion	P	p
Size	N	n

1.1 Sampling distribution

The probability distribution of a statistic over all possible samples of size n is its sampling distribution . Standard error (S.E.) = standard deviation of the sampling distribution of a statistic.

Statistic	Mean	Standard error
Sample mean x-bar	mu	sigma / sqrt(n)
Sample proportion p	P	sqrt(P Q / n)
Difference of means x1 - x2	mu1 - mu2	sqrt(s1^2/n1 + s2^2/n2)
Difference of proportions p1 - p2	P1 - P2	sqrt(P1Q1/n1 + P2Q2/n2)

Central Limit Theorem (C.L.T.)

- If x1, ..., xn is a random sample of size n >= 30 from any population with mean mu and variance sigma^2, then x-bar is approximately N(mu, sigma^2/n)
- For a normal population, x-bar is exactly normal for any n
- Basis of all large sample tests : statistic ~ N(0,1) after standardising

Finite population correction: S.E. is multiplied by sqrt((N - n)/(N - 1)) when sampling without replacement from small N.

2. Point Estimation

Estimation = using sample data to find the value of an unknown population parameter. Point estimate = a single value (e.g. x-bar for mu). Interval estimate = a range (confidence interval).

2.1 Characteristics of a good estimator

Property	Meaning
Unbiasedness	E(T) = theta. e.g. E(x-bar) = mu, E(s^2) = sigma^2 (with n - 1 divisor)
Consistency	T -> theta as n -> infinity (stochastically)
Efficiency	Among unbiased estimators, the one with least variance
Sufficiency	Contains all the information about theta present in the sample

2.2 Methods of point estimation

Method	Idea
Method of moments (Pearson)	Equate sample moments to population moments and solve
Maximum likelihood (Fisher)	Choose theta that maximises L(theta) = Product f(xi; theta)

Method	Idea
Least squares	Minimise sum of squared deviations

2.3 Common point estimates

Parameter	Estimator
Population mean mu	x-bar = Sum x / n
Population variance sigma^2	s^2 = Sum (x - x-bar)^2 / (n - 1) (unbiased)
Population proportion P	p = x / n
Poisson lambda	x-bar
Exponential theta	1 / x-bar

Example (MLE)

Poisson: L = Product e^-l l^xi / xi!. log L = -n l + (Sum xi) log l - Sum log xi!. d/dl = -n + Sum xi / l = 0 -> lambda-hat = x-bar .

3. Testing of Hypothesis

3.1 Hypotheses

Term	Meaning
Statistical hypothesis	Statement/assumption about a population parameter or distribution
Null hypothesis H0	Hypothesis of no difference / to be tested, has the equality sign (e.g. H0: mu = mu0)
Alternative hypothesis H1	Hypothesis accepted if H0 is rejected (mu != mu0, mu > mu0, mu < mu0)
Simple / composite	Specifies parameter fully / does not

- Two-tailed test: H1: mu != mu0 (difference in either direction).
- One-tailed test: H1: mu > mu0 (right tailed) or H1: mu < mu0 (left tailed).

3.2 Test statistic, critical and acceptance regions

Test statistic = quantity computed from the sample (for large samples $Z = (\text{statistic} - E(\text{statistic}))/S.E. \sim N(0,1)$). The set of values of the statistic for which H0 is rejected is the critical (rejection) region ; the remaining values form the acceptance region . The boundary values are critical values .

3.3 Level of significance

alpha = P(reject H0 when H0 is true) ; usually 5 % or 1 % . It fixes the size of the critical region.

alpha	Two-tailed	Right-tailed	Left-tailed
1 % (0.01)	$\|Z\| > 2.58$	$Z > 2.33$	$Z < -2.33$
5 % (0.05)	$\|Z\| > 1.96$	$Z > 1.645$	$Z < -1.645$
10 % (0.10)	$\|Z\| > 1.645$	$Z > 1.28$	$Z < -1.28$

3.4 Two types of errors and power

	H0 true	H0 false
Reject H0	Type I error (alpha) - producer's risk	Correct decision (power)
Accept H0	Correct decision	Type II error (beta) - consumer's risk

alpha = P(Type I error) = P(reject H0 | H0 true) beta = P(Type II error) = P(accept H0 | H0 false)

Power of the test = 1 - beta = P(reject H0 | H1 true)

For fixed n, reducing alpha increases beta . Increasing sample size reduces both. A good test has high power .

3.5 General procedure

1. State H0 and H1 → 2. Choose alpha (5 %) → 3. Select test statistic

4. Find critical value → 5. Compute Z from data → 6. Decide and conclude

Decision: if |Z calculated| > Z critical (table) -> reject H0 (significant); else accept / do not reject H0 .

4. Large Sample Tests

4.1 Test for single mean

H0: mu = mu0. Sample size n >= 30, mean x-bar:

Z = (x-bar - mu0) / (sigma / sqrt(n)) (use s if sigma unknown)

Example 1

A sample of 100 bulbs has mean life 1570 hrs, S.D. 120 hrs. Test at 5 % whether population mean = 1600 hrs. H0: mu = 1600; H1: mu != 1600. Z = (1570 - 1600)/(120/10) = -30/12 = -2.5 . |Z| = 2.5 > 1.96 -> reject H0 : the mean life differs from 1600 hrs.

Example 2 - one-tailed

A company claims average pack weight 500 g. 64 packs have mean 495 g, S.D. 16 g. Is weight less than claimed (5 %)? H1: mu < 500. Z = (495 - 500)/(16/8) = -2.5 < -1.645 -> reject H0, weight is significantly less.

4.2 Test for difference of means

H0: mu1 = mu2 (two independent large samples):

Z = (x1-bar - x2-bar) / sqrt(s1^2/n1 + s2^2/n2)

If sigma1 = sigma2 = sigma (common): S.E. = sigma sqrt(1/n1 + 1/n2).

Example 3

Two groups: n1 = 50, mean 82, S.D. 10; n2 = 60, mean 78, S.D. 12. Test equality of means at 5 %. S.E. = sqrt(100/50 + 144/60) = sqrt(2 + 2.4) = 2.098. Z = 4/2.098 = 1.907 < 1.96 -> accept H0 (difference not significant).

Example 4

Mean wages in two cities: 400 workers earn mean Rs.1500, S.D. 100 (city A); 500 workers earn Rs.1470, S.D. 120 (city B). S.E. = sqrt(10000/400 + 14400/500) = sqrt(25 + 28.8) = 7.335. Z = 30/7.335 = 4.09 > 2.58 -> highly significant, reject H0.

4.3 Test for single proportion

H0: P = P0. p = x/n = sample proportion, Q0 = 1 - P0.

$$Z = (p - P0) / \sqrt{P0\,Q0 / n}$$

Example 5

In 400 tosses of a coin 216 heads occur. Is the coin fair at 5 %? p = 216/400 = 0.54, P0 = 0.5. S.E. = sqrt(0.25/400) = 0.025. Z = 0.04/0.025 = 1.6 < 1.96 -> accept H0 : coin is fair.

Example 6

A manufacturer claims at most 5 % defective. In a sample of 200, 16 are defective. H0: P = 0.05; H1: P > 0.05. p = 0.08, S.E. = sqrt(0.05 x 0.95/200) = 0.01541. Z = 0.03/0.01541 = 1.947 > 1.645 -> reject H0 at 5 % (but accept at 1 %).

4.4 Test for difference of proportions

H0: P1 = P2. Pooled proportion P-hat = (n1 p1 + n2 p2)/(n1 + n2) = (x1 + x2)/(n1 + n2).

$$Z = (p1 - p2) / \sqrt{\text{P-hat}\ \text{Q-hat}\ (1/n1 + 1/n2)}$$

Example 7

In sample A, 20 of 200 are smokers, in sample B, 18 of 100. Test equality at 5 %. p1 = 0.10, p2 = 0.18, P-hat = 38/300 = 0.1267, Q-hat = 0.8733. S.E. = sqrt(0.1267 x 0.8733 (1/200 + 1/100)) = sqrt(0.11064 x 0.015) = 0.0407. Z = -0.08/0.0407 = -1.96 ; |Z| = 1.96 ~ critical -> borderline, reject H0 at 5 % (marginally).

Example 8

A machine A produces 40 defective out of 500, machine B 30 out of 400. P-hat = 70/900 = 0.0778; S.E. = sqrt(0.0778 x 0.9222 x (1/500 + 1/400)) = sqrt(0.07174 x 0.0045) = 0.01797. p1 = 0.08, p2 = 0.075: Z = 0.005/0.01797 = 0.278 -> accept H0; no difference between machines.

Test	H0	Test statistic Z
Single mean	mu = mu0	(x-bar - mu0)/(sigma/sqrt n)
Two means	mu1 = mu2	(x1 - x2)/sqrt(s1^2/n1 + s2^2/n2)
Single proportion	P = P0	(p - P0)/sqrt(P0Q0/n)
Two proportions	P1 = P2	(p1 - p2)/sqrt(PQ(1/n1 + 1/n2))

5. Confidence Intervals

A confidence interval (C.I.) is a range, computed from sample data, that contains the true parameter with a stated probability (confidence level 1 - alpha , e.g. 95 %). Values: 90 % -> 1.645; 95 % -> 1.96; 99 % -> 2.58 .

5.1 One-sample problems

Parameter	(1 - alpha) 100 % confidence interval
Mean mu (sigma Known / n large)	x-bar +/- Z(alpha/2) sigma/sqrt(n)
Mean mu (sigma unKnown, n >= 30)	x-bar +/- Z(alpha/2) s/sqrt(n)
Proportion P	p +/- Z(alpha/2) sqrt(p q / n)

Example 1

A sample of 100 has mean 52, S.D. 8. 95 % C.I. for mu: 52 +/- 1.96 (8/10) = 52 +/- 1.568 = (50.43, 53.57) . 99 %: 52 +/- 2.58(0.8) = (49.94, 54.06) .

Example 2 - proportion

In a survey of 400 voters 220 support a candidate. p = 0.55; S.E. = sqrt(0.55 x 0.45/400) = 0.0249. 95 % C.I. = 0.55 +/- 1.96(0.0249) = (0.501, 0.599) .

Example 3 - sample size

To estimate mean with margin of error E = 2 at 95 %, sigma = 10: n = (Z sigma/E)^2 = (1.96 x 10/2)^2 = 9.8^2 = 96.04 -> 97 .

5.2 Two-sample problems

Parameter	Confidence interval
mu1 - mu2	(x1 - x2) +/- Z(alpha/2) sqrt(s1^2/n1 + s2^2/n2)
P1 - P2	(p1 - p2) +/- Z(alpha/2) sqrt(p1q1/n1 + p2q2/n2)

Example 4

n1 = 100, x1 = 75, s1 = 10; n2 = 120, x2 = 70, s2 = 12. S.E. = sqrt(1 + 1.2) = 1.483. 95 % C.I. for mu1 - mu2 = 5 +/- 1.96(1.483) = 5 +/- 2.907 = (2.09, 7.91) . Does not contain 0 -> the means differ significantly.

Example 5 - proportions

p1 = 60/200 = 0.30, p2 = 40/150 = 0.267. S.E. = sqrt(0.3 x 0.7/200 + 0.267 x 0.733/150) = sqrt(0.00105 + 0.0013) = 0.0485. 95 % C.I. = 0.033 +/- 0.095 = (-0.062, 0.128) -> contains 0 -> no significant difference.

Relation between test and C.I.

- Two-tailed test at level alpha rejects H0: mu = mu0 iff mu0 lies outside the (1 - alpha) C.I.
- Wider interval = more confidence but less precision, larger n -> narrower interval
- Margin of error E = Z(alpha/2) x S.E.

6. Additional Concepts and Worked Problems

6.1 p-value approach

p-value = probability (under H0) of getting a result at least as extreme as the one observed. Reject H0 if p-value < alpha. For a two-tailed Z test: p = 2 P(Z > |z|).

Example: Z = 2.5 -> p = 2(0.0062) = 0.0124 < 0.05 -> reject H0 at 5 % but accept at 1 %.

6.2 Sampling distribution example

Population mean 80, S.D. 12. Sample of 36: S.E. = 12/6 = 2. P(x-bar > 83) = P(Z > 1.5) = 0.0668 . P(77 < x-bar < 83) = 2 x 0.4332 = 0.8664 .

6.3 Power of a test - example

H0: mu = 50 vs H1: mu = 52, sigma = 6, n = 36 (S.E. = 1), reject if x-bar > 51.645 (alpha = 5 %, one-tailed). If true mu = 52: beta = P(x-bar < 51.645 | mu = 52) = P(Z < -0.355) = 0.361. Power = 1 - 0.361 = 0.639 .

alpha	beta	Power
decreases	increases	decreases
n increases	decreases	increases

6.4 Steps - which test to use?

Is it a mean or a proportion? → One sample or two samples? → Is n >= 30? (large sample Z test)

Is H1 one-tailed or two-tailed? → Compare |Z| with critical value

6.5 More worked examples

Problem 1 - single mean (two-tailed)

A random sample of 900 members has mean 3.4 cm. Can it be from a population with mean 3.25 cm and S.D. 2.61 cm? $Z = (3.4 - 3.25)/(2.61/30) = 0.15/0.087 = 1.72 < 1.96$ -> accept H0: sample is from that population.

Problem 2 - proportion (one-tailed)

A dealer claims at least 60 % of customers prefer brand X. In a sample of 100, 52 prefer X. H0: $P = 0.6$, H1: $P < 0.6$. $Z = (0.52 - 0.60)/\text{sqrt}(0.6 \times 0.4/100) = -0.08/0.04899 = -1.633$. Critical (5 %) = -1.645. |Z| smaller -> accept H0 (claim not rejected).

Problem 3 - difference of means

Two samples of sizes 400 and 600 have means 85 and 80 with S.D. 15 and 20 respectively. $Z = 5/\text{sqrt}(225/400 + 400/600) = 5/\text{sqrt}(0.5625 + 0.6667) = 5/1.1087 = 4.51 > 2.58$ -> significant difference.

Problem 4 - difference of proportions

In a town A, 50 of 400 persons are illiterate, in town B, 80 of 800. $p1 = 0.125$, $p2 = 0.10$; P-hat $= 130/1200 = 0.1083$. S.E. $= \text{sqrt}(0.1083 \times 0.8917 \times (1/400 + 1/800)) = \text{sqrt}(0.09658 \times 0.00375) = 0.01903$. $Z = 0.025/0.01903 = 1.31$ -> accept H0 at 5 %.

Problem 5 - confidence interval

Sample of 64 students has mean height 168 cm, S.D. 8 cm. 95 % C.I.: $168 \pm 1.96(1) = (166.04, 169.96)$. 99 % C.I.: $168 \pm 2.58 = (165.42, 170.58)$.

Problem 6 - sample size for proportion

To estimate a proportion within 3 % at 95 % confidence (no prior estimate, $p = 0.5$): $n = Z^2\ p\ q\ /\ E^2 = (1.96)^2\ (0.25)/(0.03)^2 = 0.9604/0.0009 = 1067$.

Problem 7 - decision table

Case	Z calc	Critical	Decision
1	2.1	1.96 (two-tailed 5 %)	Reject H0
2	-1.2	-1.645 (left 5 %)	Accept H0
3	2.4	2.33 (right 1 %)	Reject H0
4	1.8	2.58 (two-tailed 1 %)	Accept H0

Formula summary	Expression
S.E. of mean	sigma / sqrt(n)
S.E. of proportion	sqrt(PQ/n)
S.E. of (x1 - x2)	sqrt(s1^2/n1 + s2^2/n2)
Pooled proportion	(n1p1 + n2p2)/(n1 + n2)
Margin of error	Z(alpha/2) x S.E.
Sample size (mean)	n = (Z sigma/E)^2

Quick revision - Unit V

- Parameter (population) vs statistic (sample); S.E. of x-bar = sigma/sqrt(n)
- H0 (equality) vs H1; alpha = P(Type I); beta = P(Type II); power = 1 - beta
- Critical Z: 1.96 / 2.58 (two-tailed 5 % / 1 %); 1.645 / 2.33 (one-tailed)
- Z tests: single mean, two means, single proportion, two proportions (pooled P)
- C.I.: estimate +/- Z(alpha/2) x S.E.